

Stock Price Trend Prediction using LSTM

Abstract

Stock market prediction is one of the most challenging tasks due to its highly volatile and non-linear nature. This project aims to predict stock price trends using a deep learning approach, specifically the Long Short-Term Memory (LSTM) neural network. Historical stock data is collected using the Yahoo Finance API and enhanced with technical indicators such as Moving Average (MA) and Relative Strength Index (RSI). The model is trained on past data to learn patterns and forecast future stock prices. The results demonstrate that LSTM models are effective in capturing temporal dependencies in time-series data and can provide reasonably accurate predictions.

Introduction

Stock price prediction plays a crucial role in financial decision-making for investors and analysts. Traditional statistical models often fail to capture complex patterns in stock data due to its dynamic and non-linear behavior. With the advancement of Artificial Intelligence and Deep Learning, models like LSTM have proven to be highly effective for time-series forecasting tasks.

LSTM is a type of Recurrent Neural Network (RNN) designed to remember long-term dependencies, making it suitable for sequential data such as stock prices. This project leverages LSTM to analyze historical stock data and predict future trends. By incorporating technical indicators like Moving Average and RSI, the model gains additional insights into market behavior, improving prediction performance.

Tools Used

- Python – Core programming language
- Pandas & NumPy – Data manipulation and numerical operations
- Matplotlib – Data visualization
- yfinance – Fetching real-time stock data
- Scikit-learn – Data preprocessing and normalization
- TensorFlow / Keras – Building and training the LSTM model
- TA (Technical Analysis Library) – Computing indicators like RSI

Steps Involved in Building the Project

1. Data Collection
Historical stock price data is fetched using the Yahoo Finance API (yfinance). The dataset includes features such as Open, High, Low, Close, and Volume.
2. Data Preprocessing
The dataset is cleaned by handling missing values and selecting relevant features. The data is normalized using MinMaxScaler to bring values into a range suitable for neural networks.
3. Feature Engineering
Additional features are created to improve model performance:
 - Moving Average (MA): Smoothens price trends
 - Relative Strength Index (RSI): Indicates overbought or oversold conditions

4. Sequence Preparation

Since LSTM requires sequential input, the data is converted into time-step sequences (e.g., past 60 days used to predict the next day).

5. Model Building

A deep learning model is constructed using Keras:

- LSTM layers to capture temporal dependencies
- Dropout layers to prevent overfitting
- Dense layers for final prediction

6. Model Training

The model is trained using historical data with a defined number of epochs and batch size. The dataset is split into training and testing sets for evaluation.

7. Prediction

The trained model is used to predict stock prices on unseen test data.

8. Visualization

Results are visualized using graphs comparing:

- Actual stock prices
- Predicted stock prices

Conclusion

This project demonstrates the effectiveness of LSTM neural networks in predicting stock price trends using historical data. By combining deep learning with technical indicators like Moving Average and RSI, the model is able to capture important patterns in stock behavior. Although stock markets are inherently unpredictable, the model provides reasonably accurate forecasts and valuable insights.

Future improvements can include adding more technical indicators, using advanced architectures like Bidirectional LSTM or Attention models, and deploying the solution as a real-time web application. Overall, this project highlights the potential of deep learning in financial time-series analysis.